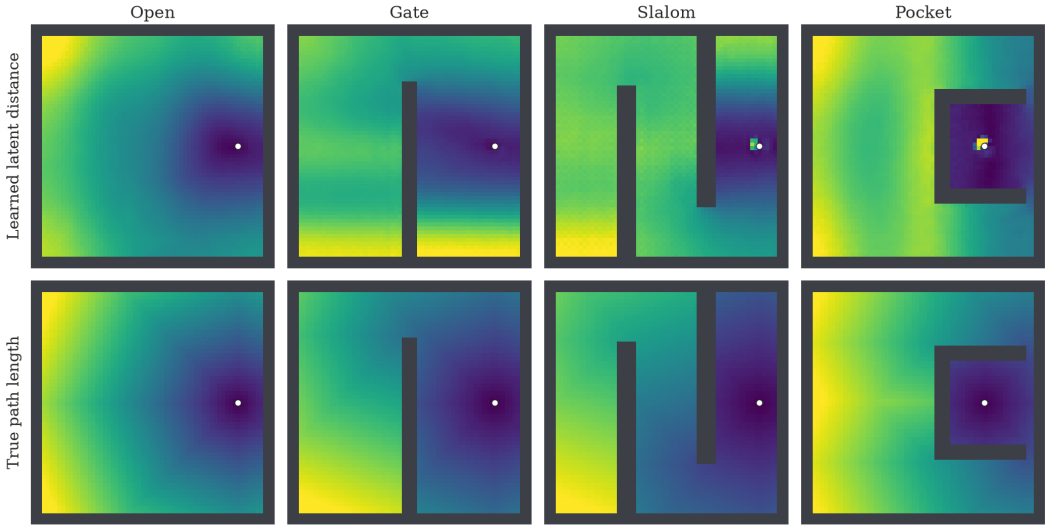
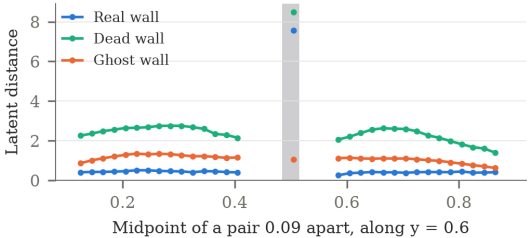
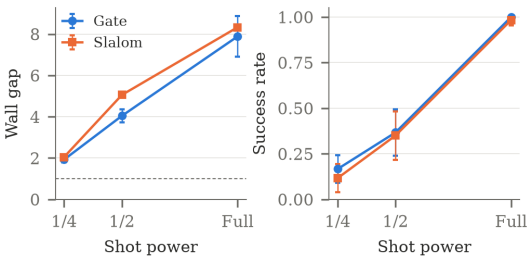
A world model that learned golf from pixels discovered the walls



A small LeWorldModel saw only random shots as 64×64 images: no rewards, no rules, no wall positions. Top row: its learned distance to the hole. Bottom row: the true path length.

Shots reveal the walls, not the picture





- A **ghost wall** is drawn in every image, but the ball rolls through it.
- Wall gap: **7.88** real, **3.82** blocks without bouncing, **0.96** ghost (1 means no wall).
- Weaker shots hit walls less, and the wall fades: 7.88, 4.04, 1.93 on Gate.
- Planning with this distance sinks 97–100% on Gate and Slalom.
- Pocket fails (48%) because of the predictor: true physics with the same cost gets 100%. My graph planner gave no clear gain.